One-pot PCR amplicon library workflow

gene-region primers + barcode primers in the same tube → pool → Nanopore (Evoprep) · HuangLab LIMS stages 1 → 3a → 3b → 4a

Stage 1 — Specimen# / Culture#

Specimen registered in LIMS
soil / environmental → Specimen# | fungal strain → Culture#

DNA extraction

Stage 3a — DNA extraction · MolWork#

DNA template → MolWork#
soil crude extract (SuperSoil) | pure fungal extraction (BOMB v9)
MolWork# minted after quantification

Soil DNA extraction
SuperSoil protocol

Fungal DNA extraction
modified BOMB v9.0

Set up one-pot PCR

One reaction tube (per sample)

gene-region primers
(head-tailed, limiting conc.)

\+

barcode primers
(barcode + head seq, excess)

template + both primer sets + polymerase master mix

Stage 3b — PCR · pcr_reaction row

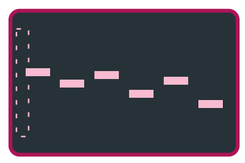
One-pot PCR
early cycles: head-tailed gene amplicon
later cycles: barcode primers take over

One-pot PCR protocol
mixture + two-phase cycling (draft v0.1)

One-pot PCR products: one tube per sample, one barcode per tube

Tube 1 barcode 1
Tube 2 barcode 2
Tube 3 barcode 3

barcode — head seq — gene — head seq — barcode



Gel check
confirm barcoded product



Qubit Flex
quantify → 10 ng/µL

conc. → equal-molar volumes

Pooling (equal molar)

Pool all barcoded samples
equal molar of every sample

LIMS register
`/lims-register`
Stage 3b PCR row → second_pcr_ok + barcode



Pooled sample

Stage 4a — Nanopore experiment · Nanopore#

Purification (bead clean-up)
PCR sample : bead = 1 : 1

Size selection 1X
bead purification protocol



Purified pooled sample

Ready to be sequenced
library construction using Evoprep

EvoPrep library prep
KAPA EvoPrep NBD114.96

Legend

- Head seq (universal tail on the gene-region primers)
- Barcode (sample-specific, on the barcode primers)
- Gene region (target amplicon; colour = different sample)

QR codes (left) open the lab's self-hosted protocol PDF for that step
left bands = HuangLab LIMS stages (/lims-register); dashed = QC / info flow
single gel check: no intermediate 1st-PCR product to inspect